RNA Gene Analysis in Cancer Patients

The goal of this project is to identify whether certain genes are expressed at different rates in the tumor tissue of breast cancer survivors versus non-survivors. To investigate this, I used the TCGAbiolinks package to download RNA-seq gene expression data. This data was collected from the TCGA Breast Invasive Carcinoma project, where cancer patients' tissue was measured for gene expression. When genes are expressed, mRNA is synthesized from the DNA and moves to the ribosome to create protein. In this data collection, they count how many RNA sequence reads that map to human genes are present in a patient's tumor cells. The data that was collected was stored in a SummarizedExperiment object that organizes the data into three components: a matrix that stores the gene expression levels in the samples, clinical data about each patient, and gene description data. The main data set used in this analysis is the gene expression levels from the various samples. Each column is a patient and each row is a gene. The value in each cell is the count of a particular RNA sequence found in that sample for that gene.

Based on my goal, I had to filter out all samples that were not tumor samples because I wanted to compare gene expression in cancerous tissue. There were also several duplicate tumor samples from the same patient, so I kept the record with the longest follow-up time, since it would have the most complete data for the patient. This is because it is the furthest recorded visit from the original diagnosis, so it is the most recent. I also separated the data into two groups: alive tumor free, and dead not tumor free. If a patient did not fit those criteria, they were filtered out because the effect the disease had on the patient was unclear. Since we have a list of patients and outcomes, we can compare the gene expression differences between the two groups.

To analyze the two groups, I used the DESeq2 package, which was created to analyze count-based gene data. To use the DESeqDataSetFromMatrix function, the gene data matrix and the patient data must match based on index. To do this, we find the common matches that exist in both data sets and order them by barcode. Genes that do not have a count of ten or more for ten or more patients are removed from the data set. I made the did-not-survive group the reference level, so all comparisons are relative to the did-not-survive group.

The DESeq2 package helps handle several challenges that occur with gene data. One is that gene data follows a negative binomial distribution more than a normal distribution. Counts also usually have high variability and are very noisy. The negative binomial distribution captures the variance being much greater than the mean in most

cases. DESeq2 does three main steps. First, it calculates a size factor for each patient to normalize the difference in sequencing depth between patients. Second, it calculates the dispersion of each gene, which measures the biological variability that exists between patients. Lastly, it uses the negative binomial model to run a Wald test on each gene to determine whether there is a statistical difference between the gene expression of survivors and non-survivors.

To complement the DESeq2 analysis, I wanted to perform a more explicit calculation that aligned more closely with the statistical methods we learned in class. I still used the size-factor-normalized gene counts generated by DESeq2 to ensure that sequencing depth differences between patients were already accounted for. I then calculated the means and dispersion for all the genes in each group. These calculations were then used to calculate the variance of a negative binomial distribution. The variance formula for this is variance = mean + (dispersion × mean^2). This formula is used because gene data counts usually have a much bigger variance than mean. After calculating these values, I wanted to perform a two-sample Z test to compare gene expression between survivors and non-survivors. However, to perform a Z test, the data must be approximately normally distributed. The raw counts are not, so to fix this, I decided to do the Z test on the log2 fold change rather than on the raw counts. Doing this transforms the data so it is approximately normally distributed, which allows a Z test to be performed. The standard error was also calculated on the log2 scale by adding the uncertainty from both groups together, and I then performed a two-sided Z test.

The DESeq2 package gave a results table that shows very significant results for many genes, including this one. The test statistic (Wald statistic) is 8.4, which is far out in the right tail and very unusual. The p-value is very small, giving very strong evidence to reject the null hypothesis.

Description: df [1 x 6]

	baseMean <dbl>	log2FoldChange <dbl>	lfcSE <dbl>	stat <dbl>	pvalue <dbl>	padj <dbl>
ENSG00000134827.8	1145.624	2.88649	0.3420353	8.43916	3.19626e-17	1.541008e-13

1 row

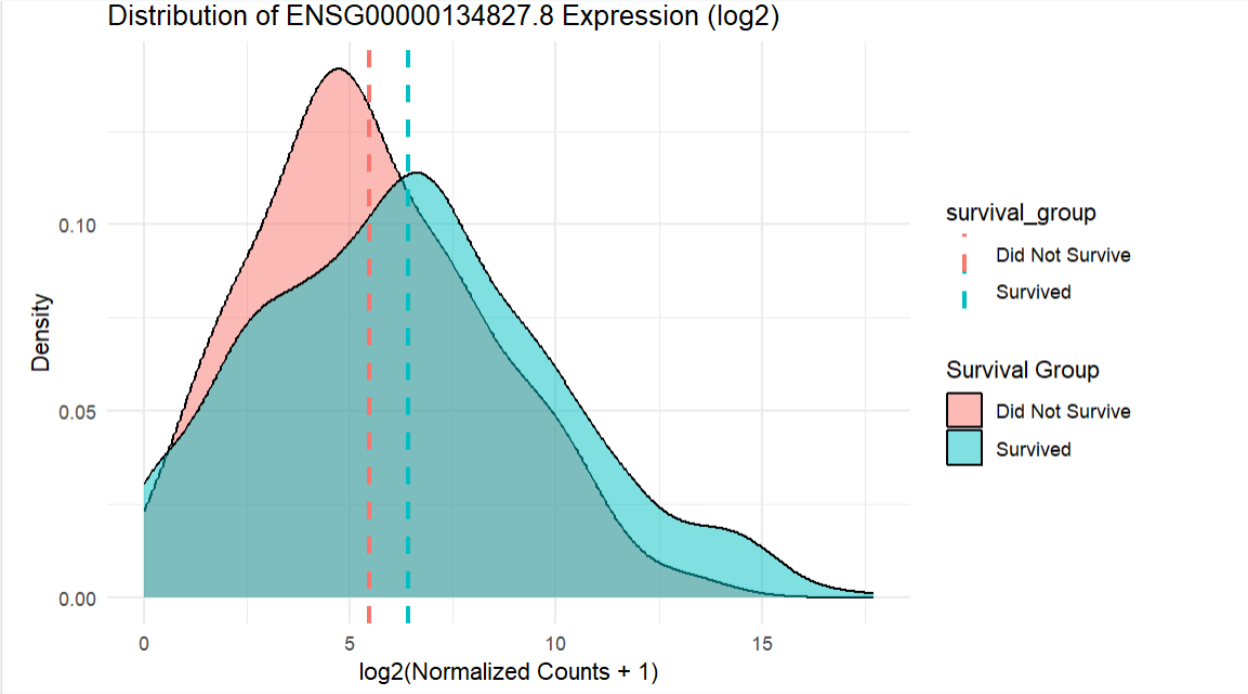
I wanted to compare those results to the more explicit technique I used, which generated similar results. I added a few more summary statistics to give more context. It is apparent that the means of the gene counts are very different between survivors and non-survivors. It also shows that survivors have higher variability in their expression of this gene than non-survivors. The Z score is still high and the p-value is very small, giving strong evidence to reject the null hypothesis.

	gene <chr>	mean_survivors <dbl>	mean_nonsurvivors <dbl>	nb_var_survivors <dbl>
ENSG00000134827.8	ENSG00000134827.8	1877.847	315.9891	100475034

1 row | 1-5 of 7 columns

nb_var_survivors <dbl>	nb_var_nonsurvivors <dbl>	z_approx <dbl>	pval_approx <dbl>
100475034	1244012	4.053637	5.042743e-05

To show the difference more visually, I made a graph of the log2 transform of the gene counts. It shows the shift in the mean and distribution between the two groups.



I wanted to do a power test to investigate how often the difference between the two groups would be detected, so I did a two-sample t test power calculation. For N, I used the effective sample size since the groups had different sample sizes. Based on this test, the difference would be detected 91% of the time using a significance level of 0.05.

```
     Two-sample t test power calculation

              n = 142.8738
          delta = 2.571132
             sd = 6.528247
      sig.level = 0.05
          power = 0.9126982
    alternative = two.sided

NOTE: n is number in *each* group
```

I thought it would also be interesting to find what sample size would be needed to reach a power of 99%. To do this, I just had to set n = NULL and set the power to 0.99. It was concluded that a sample size of 238 would be needed to accomplish this.

```
     Two-sample t test power calculation

              n = 237.8525
          delta = 2.571132
             sd = 6.528247
      sig.level = 0.05
          power = 0.99
    alternative = two.sided

NOTE: n is number in *each* group
```

I wanted to see whether any previous research validated my finding that higher expression of the gene TCN1 in breast cancer tissue is linked to better patient outcomes. I found a study with similar findings titled “Oxeiptosis gene expression profiling identified TCN1 as a prognostic factor for breast cancer.” One of their data sources was the same TCGA-BRCA, but they used some others as well. They found that cancer patients whose tumors expressed TCN1 at a higher level had an increased survival rate, with a p-value of 0.00046. One reason they mention for why this might be the case is that the TCN1 gene can

help promote the oxidative stress response pathway. This pathway is responsible for killing the cell, and promoting this cell death pathway hinders the replication of tumor cells. They also found that increased TCN1 activated the immune system more against the tumor, so cancer cells cannot hide as well. Overall, I found this statistical analysis very interesting and may continue it in more detail in the future. I think it would be interesting to create a logistic regression model that uses highly expressed genes associated with better patient outcomes to predict survival.

Sources

Zhu, Y., Zang, L., Zeng, D., Zhou, P., and Hu, J. (2025). Oxeiptosis gene expression profiling identified TCN1 as a prognostic factor for breast cancer. Oncologie, 27(1), 95–109. https://doi.org/10.1515/oncologie-2024-0397

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https://bioconductor.org/packages/release/bioc/html/TCGAbiolinks.html

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